ModelIt!

Student Activity Pack

Your Body Is a City of Trillions

How Your Organ Systems Work Together to Keep You Alive

Grade 6 | NGSS MS-LS1-3 | Systems and System Models

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What happens to your breathing and heartbeat when you run?

2 Why do you need to eat food? What does your body do with it?

3 How does the oxygen you breathe get to your muscles?

4 Draw the path food takes through your body from mouth to muscle.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Heart Rate | Blood Oxygen Level | Nutrient Intake | Physical Activity

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Resting State

Low physical activity, normal nutrients and oxygen

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Sprint Mode

Maximum physical activity

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Skipping Lunch

Low nutrients, moderate activity

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Body Temperature — Internal temperature rises during exercise; the body must cool itself through sweating and blood flow to skin

Lactic Acid Buildup — During intense activity, muscles produce lactic acid faster than the body can clear it, causing fatigue and burning

Hydration Level — Water is essential for blood volume, sweating, and chemical reactions — dehydration degrades all body systems

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How does your body run a city more complex than any humans have ever built — 24/7, without you even thinking about it? Use evidence from your simulation to support your answers.

1 Why does your heart beat faster when you exercise? What problem is it solving?

2 What would happen to your muscles if your circulatory system stopped working?

3 How is your body like a city? Give at least three specific comparisons.

4 Why do athletes need to eat differently than people who sit at desks all day?

5 How does homeostasis keep your body working even when conditions change?

8 STEM Challenge: Design an Athletic Performance Plan

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YOUR ENGINEERING MISSION

Design an evidence-based training and nutrition plan for a student athlete that optimizes how their body systems work together during competition.

THE SCENARIO

A middle school track team needs help. Their coach wants your biomedical team to design a plan that helps runners perform their best in the 400-meter race. Use your body systems model to explain your recommendations.

Engineering Design Thinking

1. WHAT FOODS WILL PROVIDE THE RIGHT TYPE AND AMOUNT OF ENERGY FOR A SPRINT?
2. HOW SHOULD THE ATHLETE WARM UP TO PREPARE THEIR CIRCULATORY AND RESPIRATORY SYSTEMS?
3. WHAT IS THE IDEAL HEART RATE ZONE FOR PEAK PERFORMANCE VS. RISK OF EXHAUSTION?
4. HOW WILL YOU MEASURE WHETHER YOUR PLAN ACTUALLY IMPROVES PERFORMANCE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Exercise Physiologists and Sports Medicine Specialists study how the body responds to physical activity and design training programs for athletes and patients. They earn $55,000–$100,000/year.